Architecture Justification - Digital Campus Library Reservation Gateway

Kushal G | PES1UG24AM145 | Section C | Team 08 - Problem Statement #08

Architecture Selection

We chose Layered Architecture for the Digital Campus Library Reservation Gateway (DCLG). The Web Portal UI forms the Presentation layer; Catalog, Hold & Reservation, Circulation, Fine Management & Payment, and Notification services form the Business layer; and a controlled Data Access & Security component protects the shared Library Database in the Data layer. The services expose clear interfaces so each responsibility can be tested and changed without coupling the UI directly to storage.

Architectural Style Comparison

Style	Strength for DCLG	Main limitation	Fit
Layered	Clear separation; central transactions help keep fine balances and FIFO holds consistent.	The shared data tier needs careful tuning and high availability.	High
Client-Server	Simple deployment and centralized control.	A single server can bottleneck search or become a single point of failure; UI and rules can become coupled.	Low
Microservices	Independent scaling and fault isolation.	More deployments, network calls and distributed consistency work for holds, loans and fine-ledger updates.	Medium

Two Reasons for the Choice

1. Consistent library transactions. FIFO hold queues, the 48-hour collection window, loan status, overdue-fine accrual and payment-ledger updates depend on consistent shared state. A controlled business layer and common data-access path make it easier to apply validation and transactional updates together than to coordinate several independently owned service databases.

2. Appropriate scale with room to meet the NFRs. DCLG serves one campus and a catalog of 50,000 titles. Separating catalog search, holds, circulation, fines and notifications keeps responsibilities modular without introducing the operational overhead of full microservices. Indexed ISBN lookups and pagination support the under-150-ms search target; replicated stateless application nodes and monitored database failover are proposed to support 99.9% availability and recovery from a single-node failure within five minutes.

Security Advantage

The Web Portal never connects directly to the Library Database. The business services validate the authenticated member and enforce role-based permissions: students can view and pay only their own fines, while catalog changes are restricted to the Head Librarian. Fine Management sends payment requests through the external payment gateway; the library stores the payment outcome/reference rather than raw card details.

Performance Benefit

Catalog Service uses indexed, paginated ISBN queries against the 50,000-title repository, avoiding unnecessary service-to-service network hops on the search path. Hold-ready and receipt messages are dispatched through the Notification Service outside the main check-in/payment response, helping the system stay responsive while meeting the five-minute notification requirement.

Trade-off and Mitigation

Layered architecture concentrates data access, so the database tier can become a bottleneck. Mitigate this with indexed queries, connection pooling, monitored failover, backups and replicated application nodes; these are design measures, not claims of measured test results.

Digital Campus Library Reservation Gateway (DCLG)

UML Component Diagram - Layered Architecture | SE Lab 3 - Team 08

